

Conversions



1 Convert the following measurements:

- | | |
|---------------------------|--------------------------|
| a 1 cm to millimetres. | b 3 cm to millimetres. |
| c 4.65 cm to millimetres. | d 60 mm to centimetres. |
| e 88 mm to centimetres. | f 1 m to centimetres. |
| g 6 m to centimetres. | h 7.33 m to centimetres. |
| i 800 cm to metres. | j 14.9 cm to metres. |
| k 13 898 cm to metres. | l 9000 m to kilometres. |
| m 12 226 m to kilometres. | n 1 km to metres. |
| o 6 km to metres. | p 6.69 km to metres. |

2 Convert the following measurements:

- | | |
|-----------------------------|-------------------------------|
| a 1 m to millimetres. | b 14 m to millimetres. |
| c 9.03 m to millimetres. | d 3484 mm to metres. |
| e 6 km to centimetres. | f 7.43 km to centimetres. |
| g 866 800 cm to kilometres. | h 9 km to millimetres. |
| i 6.71 km to millimetres. | j 1 528 400 mm to kilometres. |

Applications

3 A desk is 1.39 m long. A rug is 120 cm long.

- a Convert the length of the rug from centimetres to metres.
- b State whether the desk or the rug is longer.

4 Purnjay is 176 cm tall. Nadia is 1.51 m tall.

- a Convert Purnjay's height from centimetres to metres.
- b Which person is shorter?

5 State which person is taller in the following pairs:

- a Haonan is 165 cm tall and Tina is 1.53 m tall.
- b Hermione is 1.23 m tall and Sandy is 161 cm tall.
- c Joanne is 127 cm tall and Ellie is 1.41 m tall.



6 A coffee table is 1.17 m long. A whiteboard  147 cm long.

- a Convert the length of the coffee table from metres to centimetres.
- b State whether the coffee table or the whiteboard is longer.

7 Plant 1 grows 3.6 cm in a week. Plant 2 grows 39 mm in the same week.

Which plant grows more in the week?

8 A rectangular table is measured to be 586 centimetres long and 3.64 metres wide.

- a Convert the length of the rectangular table to metres.
- b Calculate the difference between the length and the width in metres.

9 A rectangular table is measured to be 5.23 metres long and 4.99 metres wide. Calculate the difference in metres between the length and the width of the table.

10 The Qinghai-Tibet railway in China includes the Tanggula railway station. At 5068 metres high, it is the world's highest railway station.

- a How many centimetres high is the station?
- b How many kilometres high is the station?

11 If a fly is 1.16 cm long and a bee is 19 mm long, how much longer is the bee than the fly in centimetres?

12 If a mosquito is 1.23 cm long and a wasp is 13 mm long, how much longer is the mosquito than the wasp in millimetres?

13 A tower is 125 metres tall.

- a Find the height of the building in kilometres.
- b Find the height of the building in centimetres.

14 The following table shows the distance from New York to other cities in the United States:

City	Distance (km)
Los Angeles	3940
Houston	2284
Washington	328
Chicago	1146
Charlotte	856

- a Which city is furthest from New York?
- b Which city is closest to New York?
- c Which city is about 1000 km further away from New York than Charlotte?



City	Distance (km)
New Orleans	1882

15 A car is 4 m long. A truck is 934 cm long.

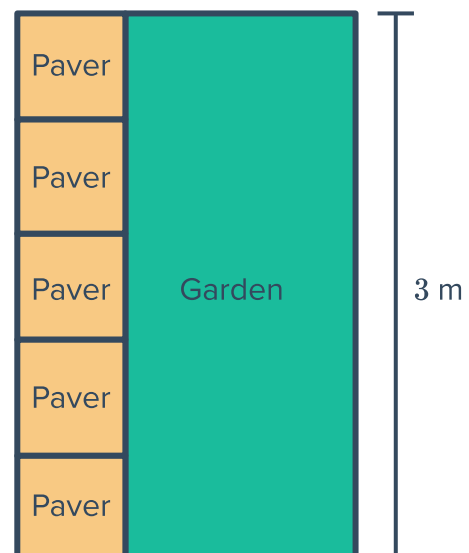
- a** Find the length of the truck in metres.
- b** How many metres longer is the truck than the car?

16 A sail boat is 6 m long. A yacht is 956 cm long.

- a** Find the length of the sail boat in centimetres.
- b** How many centimetres longer is the yacht than the sail boat?

17 Michael wants to lay a row of square pavers along the edge of his 3 m garden bed as shown in the picture:

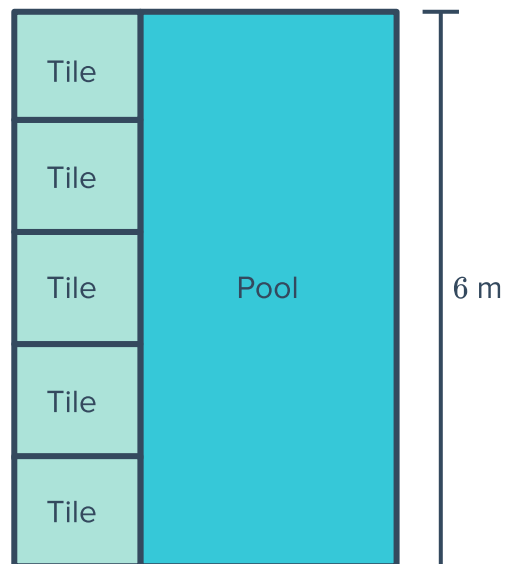
Find the side length of each paver in metres.



- 18 Ryan wants to lay a row of square tiles along the edge of his 6 m pool as shown in the picture:



Find the side length of each tile in metres.



- 19 The table below shows the different skyscrapers around the world and their heights:

Name	Height (meters)	Height (kilometers)
One World Trade Center	541.3	
Shanghai Tower		0.632
Petronas Twin Towers	451.9	
Zifeng Tower		0.45
Burj Khalifa	828	

- a Complete the conversions for the given skyscrapers in the table above.
- b Which skyscraper is the tallest?
- 20 A brick is 230 mm long.
- a Find the length of a single brick in centimetres.
- b If 19 bricks were lined up lengthways to build a fence, how long would the fence be? Give your answer in centimetres.
- c Find the length of the fence in metres.

21 The wingspan of a jumbo jet is 6000 cm:



- a** If 21 jumbo jets are lined up wing-to-wing, find the length of the total wingspan in centimetres.
- b** Convert this length to kilometres.

